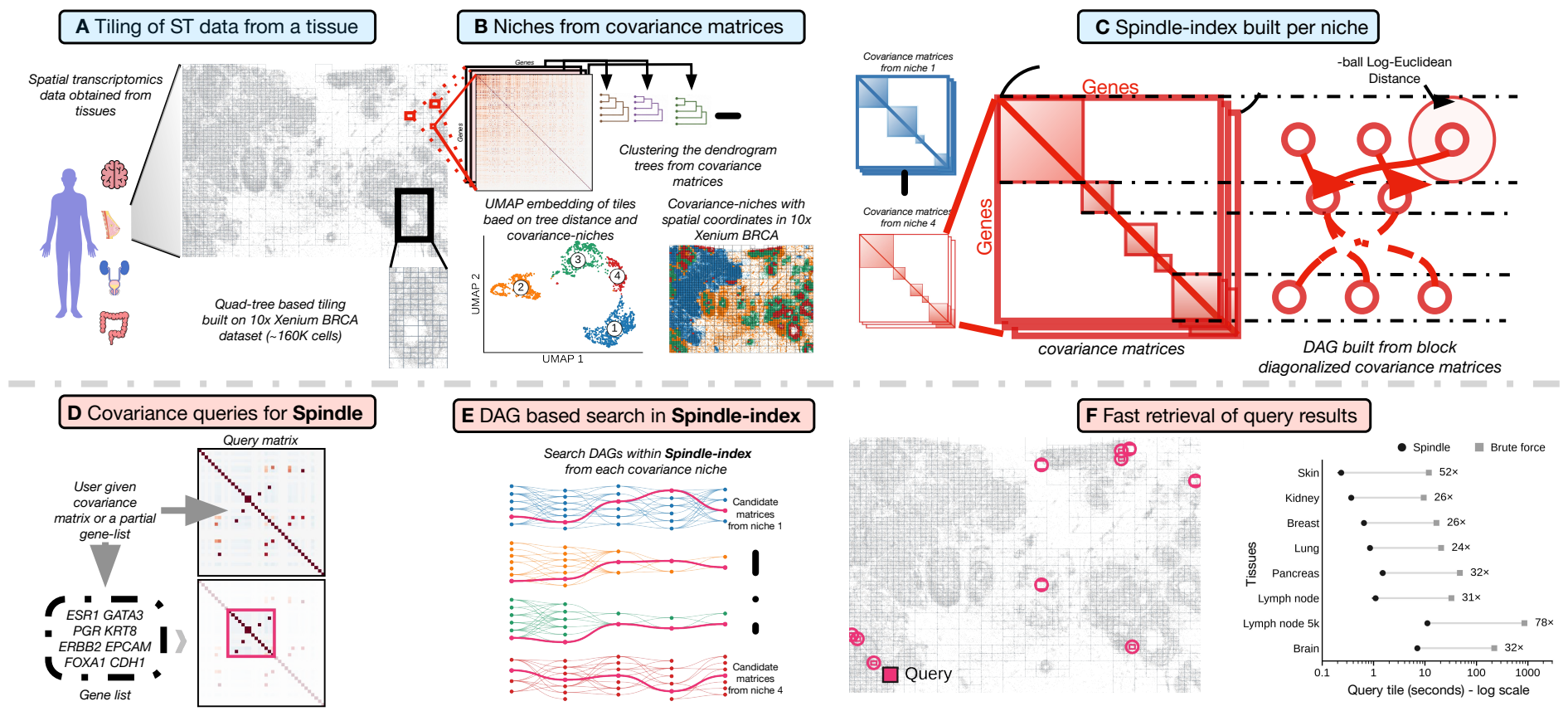

A Tiling of ST data from a tissue
Spatial transcriptomics data obtained from tissues
Quad-tree based tiling built on 10x Xenium BRCA dataset (~160K cells)
B Niches from covariance matrices
Genes
Clustering the dendrogram trees from covariance matrices
UMAP embedding of tiles baed on tree distance and covariance-niches
UMAP 2
UMAP 1
Covariance-niches with spatial coordinates in 10x Xenium BRCA
C Spindle-index built per niche
Covariance matrices from niche 1
Covariance matrices from niche 4
Genes
-ball Log-Euclidean Distance
covariance matrices
DAG built from block diagonalized covariance matrices
D Covariance queries for Spindle
Query matrix
User given covariance matrix or a partial gene-list
ESR1 GATA3 PGR KRT8 ERBB2 EPCAM FOXA1 CDH1
Gene list
E DAG based search in Spindle-index
Search DAGs within Spindle-index from each covariance niche
Candidate matrices from niche 1
Candidate matrices from niche 4
F Fast retrieval of query results
Query
Spindle
Brute force
Skin 52×
Kidney 26×
Breast 26×
Lung 24×
Pancreas 32×
Lymph node 31×
Lymph node 5k 78×
Brain 32×
Tissues
0.1
1
10
100
1000
Query tile (seconds) - log scale